Huawei Access Node Routing Protocol Flap Resolution Procedure for Malleswaram Node

Node: MALL-ACC-03 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_8831

Description

The routing protocol flap failure domain, denoted by alarm code ALARM_8831 with a severity of Major, occurs on the Huawei Access Node (MALL-ACC-03) situated at the Malleswaram Node. This failure domain is precipitated by a multitude of factors, including but not limited to, BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. These phenomena are often the result of underlying issues such as network topology changes, equipment misconfigurations, or hardware malfunctions. In the context of the Huawei Access Node, the routing protocol flap can lead to network instability, packet loss, and ultimately, service disruption. The Huawei CLI dialect is utilized to diagnose and resolve these issues, leveraging commands that scrutinize the control and data planes to identify and rectify the root cause of the failure. The intricacies of BGP, OSPF, and ISIS protocols, alongside the specific hardware and software configurations of the Huawei equipment, necessitate a nuanced and meticulous approach to troubleshooting and resolution. By understanding the complexities of these protocols and the Huawei CLI, network engineers can effectively mitigate the routing protocol flap, ensuring the stability and reliability of the network infrastructure.

Pre-requisite Checks

- Dispatch remote hands to Malleswaram Node for physical access
- Verify redundant link status to ensure no single point of failure
- Check environmental conditions (temperature, power) at the node

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp peer, Expected Output: BGP peer information, including state and statistics, Description: This command isolates issues in the BGP control plane, providing insight into peer relationships and potential route churn causes.

Command: display ospf interface, Expected Output: OSPF interface information, including adjacency status, Description: This command examines the OSPF data plane, focusing on interface configurations and adjacency issues that could lead to routing protocol flaps.
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Command: display isis interface, Expected Output: ISIS interface information, including LSP flooding details, Description: This command delves into the ISIS protocol specifics, particularly the interface configurations and LSP flooding behaviors, to identify potential causes of routing instability.
```

Resolution Steps

Step 1: Step: Adjust BGP Peer Configuration, Command: bgp peer description , Description: Modify BGP peer configurations to stabilize route exchange and mitigate churn, ensuring accurate and consistent route information.

Step 2: Step: Restore OSPF Adjacency, Command: ospf area interface , Description: Reestablish OSPF adjacencies by adjusting interface and area configurations, ensuring proper neighbor discovery and route propagation.

Step 3: Step: Optimize ISIS LSP Flooding, Command: isis lsp-interval , Description: Tune ISIS LSP flooding parameters to reduce unnecessary updates, thereby stabilizing the routing table and minimizing protocol flaps.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Stable BGP routing table, Description: Verify the stability of the BGP routing table post-configuration adjustments, ensuring no ongoing route churn.

Command: display ospf routing-table, Expected Output: OSPF routing table with stable adjacencies, Description: Confirm the stability of OSPF adjacencies and the accuracy of the routing table, indicating successful restoration of neighbor relationships.

Command: display isis routing-table, Expected Output: ISIS routing table with minimized LSP flooding, Description: Validate the optimization of ISIS LSP flooding, ensuring reduced protocol overhead and a stable routing environment.
```

Roll-back Procedures

- Step: Revert BGP Peer Configuration, Command: undo bgp peer , Description: Revert BGP peer configurations to their pre-adjustment state, should the initial fix introduce unforeseen issues or instability.

- Step: Restore Original OSPF Configuration, Command: undo ospf area interface , Description: Reestablish the original OSPF configurations, should adjacency restoration attempts fail or introduce new problems.

- Step: Reset ISIS LSP Flooding Parameters, Command: undo isis lsp-interval, Description: Reset ISIS LSP flooding parameters to their default or previous settings, in case optimization attempts lead to unforeseen consequences.